

The sun's energy

In the tropics, near the equator, the sun's rays strike Earth at a steep angle, so the energy is very concentrated. But near the poles, sunlight hits the surface at a narrow angle. This spreads the sun's energy, giving a weak heating effect. The result is that polar regions are much colder than tropical zones, allowing ice to form in the Arctic and Antarctic. The difference in the solar heating at different latitudes sets bodies of air and seawater in motion, driving winds and ocean currents.

High latitudes (near the poles) receive sunlight at a low angle, dispersing its heat energy over a wider area than in the tropics

The tropics (near the equator) receive sunlight at a steep angle, so the heat is focused onto a smaller area than at the poles

